

Cash Got Your Tongue?

CEO Language-Uncertainty

Around Undisclosed Cash Acquisitions

by

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Abstract

Firms host quarterly earnings calls and field analysts' unscripted questions, and the language a chief executive uses in answering is increasingly read as a window into what they know but have not yet disclosed. When a firm is privately committed to an acquisition it has not yet announced, the executive holds the routine call unable to confirm or deny the one development that matters, and whether the spoken record tracks that withholding has not been characterized. We study United States public firms' earnings-call transcripts and acquisition records from 2002 to 2018. From this language we construct a residual measure of chief-executive question-and-answer uncertainty, the component that remains once each executive's persistent speaking style is removed, and we track it around firms' cash acquisitions against stock acquisitions as a placebo. This residual uncertainty is elevated in the quarter before a cash acquisition, relative to the firm's own other quarters, with no comparable rise before stock acquisitions. It becomes indistinguishable from zero once the deal is announced, even before completion, while the cash that funds the purchase persists on the balance sheet until the deal closes. The elevation concentrates in cash acquisitions rather than stock acquisitions, a difference that survives a formal pooled test. This pattern is not explained by analysts devoting more of the call to cash questions. The residual is also unrelated to the firm's post-call bid-ask spread, a standard gauge of information asymmetry, while the scripted presentation is positively associated with it, consistent with the two parts of the call relating to the outside information environment in different ways. We read these patterns as call language tracking the deal's passage from private to public, a within-firm correlational regularity rather than a tested mechanism.

Keywords: earnings conference calls; CEO speech uncertainty; mergers and acquisitions; information asymmetry; corporate cash holdings.

JEL Classification: D82; G14; G32; G34; M41.

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Dedication

For the mothers who danced at their children's graves in mournful defiance.

This effort is dedicated to the tens of thousands murdered in January 2026, the largest massacre in modern Iranian history.

Your dreams were taken, may your light shine upon the path ahead.

Table of Contents

List of Tables	ix
1 Introduction	1
2 Conceptual Framework and Empirical Strategy	4
2.1 Conceptual Framework	4
2.2 Hypothesis Development	8
2.3 Estimation of the Main Variable	10
2.4 Methodology and Empirical Design	12
2.5 Specification and Measurement of Key Constructs	15
3 Main Empirical Analyses	17
3.1 Data, Sample, and Variable Construction	17
3.2 Main Analysis 1: The Pre-Announcement Run-Up	20
3.3 Main Analysis 2: Differential Timing Around the Announcement	22
3.4 Main Analysis 3: Cash-Specificity	24
4 Additional Analyses	26
4.1 Ruling Out Analyst Scrutiny	26
4.2 Outsider Reactions: The Bid-Ask Spread	28
4.3 Robustness: Withdrawal as a Resolution Event	30
4.4 Robustness: The Cash Result Without the Dynamic Term	31

5 Conclusion	32
References	34
Appendices	51

List of Tables

5.1	Summary Statistics	36
5.2	Pre-Announcement Run-Up Test: Cash versus Stock Acquirers	37
5.3	Uncertainty and Cash Around the Announcement (Matched Sample)	38
5.4	Event Study by Payment Type: Cash versus Stock Acquirers	39
5.5	Cash-Specificity Test: Cash versus Stock Acquirers	40
5.6	Convergent Validity: Firm Political Risk	41
5.7	Convergent Validity: US Policy Uncertainty	42
5.8	Convergent Validity: Global Policy Uncertainty	43
5.9	Analyst Cash Scrutiny: Construct Validity	44
5.10	Analyst Cash Scrutiny as a Driver of CEO Uncertainty	45
5.11	Analyst Cash Scrutiny: Interaction with the Run-Up	46
5.12	CEO Uncertainty and the Bid-Ask Spread	47
5.13	Robustness: Withdrawal as a Resolution Event	48
5.14	Robustness: Static Fixed Effects without the Lagged Dependent Variable	49
5.15	Construct Validity: Replication of Dzieliński et al. (2021)	50

Chapter 1

Introduction

Earnings calls, and especially the unscripted question-and-answer session, are a setting in which managers' language carries information beyond the scripted presentation (Matsumoto et al., 2011). Disclosure theory holds that an informed manager may rationally withhold private information when disclosure is costly (Verrecchia, 1983), and that non-disclosure can persist in equilibrium because outside investors cannot distinguish an uninformed manager from one who is informed but silent (Dye, 1985), so a withholding state is itself informative. A firm privately committed to an acquisition it has not announced sits in a concrete such state: information about preliminary merger negotiations can be material well before any definitive agreement (Basic Inc. v. Levinson, 1988), and although the firm may stay silent it may not mislead once it speaks (Rule 10b-5, 2014; Basic Inc. v. Levinson, 1988), so the executive can neither confirm nor deny the pending deal and is left to answer around it. When a manager cannot answer directly, the constraint can surface in the words themselves, for non-answers are common and not informationally empty: in a large share of calls managers leave requested information undisclosed, and, in the words of that literature, silence speaks (Hollander et al., 2010).

Reading this language matters because the information in an undisclosed acquisition is exactly what securities law keeps off the call, yet the unscripted question-and-answer session is one of the few recurring venues in which a constrained executive must keep speaking and fielding questions. If the bind leaves a trace in the words, that trace would mark a deal's passage from private to public in the firm's own voice, in a channel separate from the price run-up already documented around mergers (Keown and Pinkerton, 1981).

The nearest prior work reads deal-related language along other dimensions. Thewissen et al. (2024) show that acquirers manage the tone of their disclosure before stock-for-stock deals, and Ragozzino and Reuer (2024) find that the volume of corporate-strategy vocabulary rises on the

calls of acquisitive firms; both concern tone or subject matter rather than expressed uncertainty. Separately, Keown and Pinkerton (1981) document that much of a merger's price reaction precedes its public announcement, an anticipatory signal that nonetheless lives in prices rather than in language. To our knowledge no prior work reads uncertainty language in the unscripted question-and-answer session in the anticipatory window before a withheld deal; we offer this as a positioning claim about where the contribution sits, not as a tested mechanism.

The purpose of this study is to ask whether chief-executive question-and-answer uncertainty tracks a deal's passage from private to public: whether it is elevated while an acquisition is withheld and recedes once it is announced. We construct a residual measure of chief-executive question-and-answer uncertainty, *UncResCEO*, the call-varying component that remains once each executive's persistent speaking style is netted out, following the earnings-call language decomposition of Dzielinski et al. (2021), and we defer its construction and the estimating equations to Section 2. We study how this measure behaves around cash acquisitions, using cash deals against a stock-acquisition placebo that shares the same withholding setting, on United States public, non-financial, non-utility firms' earnings calls from 2002 to 2018.

Our main findings can be previewed as follows. First, residual chief-executive uncertainty is elevated in the quarter before a cash acquisition, relative to the same firm's other quarters, with no comparable rise before stock acquisitions. Second, this elevation is anticipatory: it becomes indistinguishable from zero as soon as the deal is announced, even before it closes, whereas the cash that funds the purchase persists on the balance sheet until completion, two paths running on different clocks. This round-trip contrast is the study's strongest result.

Finally, the run-up concentrates in cash acquisitions rather than stock acquisitions, a difference that survives a formal pooled test of cash-specificity, though we read it as concentration rather than strict specificity, and the cash-accumulation mechanism behind it is left open. We also rule out the most immediate alternative. The run-up is not accounted for by analysts devoting more of the call to cash questions: with a direct measure of analyst cash scrutiny and its interaction entered alongside the pre-announcement indicator, the elevation survives and scrutiny does not amplify it. This means that scrutiny does not account for this particular run-up, not that scrutiny never matters. A further analysis turns to the firm's post-call information environment, measured by the bid-ask spread: the residual is unrelated to it, while the scripted presentation is positively associated with it, a pattern consistent with the scripted and unscripted segments reaching the outside environment in different ways.

This study makes four contributions, each descriptive. First, it reads a residual measure of chief-executive question-and-answer uncertainty, the call-varying component that remains once each executive's persistent speaking style is netted out, in the anticipatory window before an acquisition is public, where the nearest prior work has read the other dimensions of deal-related

language catalogued above rather than uncertainty language. Second, it documents that this language pattern concentrates in cash acquisitions rather than stock, a contrast that survives a formal pooled test. Third, it reads the unscripted question-and-answer session as tracking a deal's passage from private to public, connecting the literature on when managers withhold private information with the evidence that markets anticipate corporate transactions, while claiming no causal channel. Fourth, it documents that this residual uncertainty is unrelated to the firm's post-call bid-ask spread while the scripted presentation is positively associated with it, consistent with a difference between the scripted and unscripted segments in how they relate to the outside information environment.

Throughout, we read these patterns as correlational and within-firm. The design supports no causal identification, establishes no mechanism, and leaves open why uncertainty language concentrates in cash deals, with a compliance-constrained inability to speak and a strategically chosen reticence remaining observationally equivalent.

The remainder of the paper proceeds as follows. Section 2 develops the conceptual framework and empirical strategy; Section 3 presents the three main analyses, the pre-announcement run-up, the differential timing around announcement, and the cash-specificity test; Section 4 reports additional analyses, ruling out analyst scrutiny, examining the post-call bid-ask spread, and checking the robustness of the differential-timing result; and Section 5 concludes.

Chapter 2

Conceptual Framework and Empirical Strategy

2.1 Conceptual Framework

A firm that has privately committed to an acquisition but not yet announced it occupies a distinct *disclosure state*. A pending acquisition that is material to the acquirer constitutes material nonpublic information: information about preliminary merger negotiations can be material well before any definitive agreement, with materiality determined by balancing the probability that the transaction is consummated against its magnitude to the firm (Basic Inc. v. Levinson, 1988). Securities law imposes no general duty to disclose such confidential negotiations, so silence is permitted; but once a firm speaks it may not mislead, since an untrue or half-true statement is unlawful (Rule 10b-5, 2014; Basic Inc. v. Levinson, 1988). The quarterly earnings call is a recurring event that managers nonetheless hold, so the executive faces live questioning while unable to address the one material development and unable to deny it. Disclosure theory tells us such a withholding state is not informationally empty: an informed manager may rationally withhold private information when disclosure is costly, sustaining a threshold below which information is withheld (Verrecchia, 1983), and non-disclosure persists in equilibrium precisely because outside investors cannot distinguish a manager who is uninformed from one who is informed but silent (Dye, 1985). This withholding bind—material information held back while the firm keeps fielding questions—is the organizing primitive of our analysis, which we read in descriptive, correlational terms throughout.

The venue in which this bind becomes observable is the quarterly earnings conference call, which pairs a scripted managerial presentation with an unscripted question-and-answer session

in which analysts question management directly. The two segments are not equally revealing: both are incrementally informative relative to the accompanying press release, and the discussion carries information content beyond the presentation (Matsumoto et al., 2011)—making the unscripted exchange, which managers cannot fully prepare in advance, the natural place to observe a constrained executive under questioning. That language can be read quantitatively: because general-purpose word lists misclassify financial text—nearly three-quarters of the Harvard dictionary’s negative-word counts are attributable to words that are typically not negative in a financial context—Loughran and McDonald (2011) construct dictionaries specific to financial disclosure, among them an uncertainty list of 285 words organized around imprecision rather than risk. We adopt this established apparatus—the presentation/Q&A split and the finance-specific dictionary, including its uncertainty list—to read the language of the call.

Within this venue the bind has an observable consequence. Because the pending acquisition is material and nonpublic, the executive can neither address the analysts’ questions that bear on it nor deny that anything is afoot, so a direct answer is unavailable. Such non-answers are common, and they are not informationally empty: in roughly six of ten calls managers withhold requested information, leaving questions unanswered, and, in the words of that literature, silence speaks (Hollander et al., 2010). But silence is not a costless refuge—precisely because it speaks—and the setting is an unscripted exchange rather than prepared remarks. Non-disclosure therefore need not surface as outright silence; it can surface in the words themselves, as answers that grow hedged, qualified, and non-committal—elevated uncertainty language in the Q&A. We read this language, rather than the incidence of silence, because the constraint binds while the executive keeps speaking. The resulting pattern is keyed to the disclosure state: the bind predicts that uncertainty language is elevated while the acquisition remains undisclosed and recedes once announcement makes it public—a directional prediction we test rather than assume. This anticipatory timing—present before the announcement, absent after—is the first of the two dimensions we develop.

The anticipatory pattern just described, however, is not the only thing moving uncertainty language on a call. How uncertain a CEO sounds is partly a durable personal trait: some executives simply speak more cautiously than others, quarter after quarter. Bertrand and Schoar (2003) show that managers carry such persistent, individual styles—manager fixed effects—that durably shape the policies of the firms they run, with a manager’s imprint in one firm strongly predicting his imprint in the next across investment, financial, and organizational policy alike. Dzielinski et al. (2021) construct such a decomposition for the language of earnings calls, separating a CEO’s use of uncertainty words into a persistent, manager-specific style component and a call-varying residual. The implication for our purpose is a locating one. A persistent style is a fixed trait of the speaker, not a signal that tracks a particular, still-undisclosed transaction; an anticipatory component, by contrast, must emerge while a given deal is pending and subside once it is announced—it must vary from call to call within a single executive’s tenure. If such a component exists, it therefore

cannot reside in the persistent style; it can only live in the call-varying residual that remains once persistent style is netted out. This locates where an anticipatory signal must be—it does not claim the residual is that signal, which only the research design that follows can isolate.

A second dimension concerns not the timing of the signal but the deals in which it should concentrate. Cash and stock acquisitions share the identical disclosure setting—both are material, both are withheld—yet differ in one design-relevant respect: a cash purchase draws on an accumulated cash position, whereas a stock exchange need not. Harford (1999) documents the underlying fact: firms accumulate cash reserves above a baseline model of normal holdings, and cash-rich firms are more likely to make acquisitions. That accumulation is a by-product of retained free cash flow rather than saving toward a particular deal, so we take from it only that a cash bid draws on a pre-existing cash position—not that firms stockpile in order to fund planned acquisitions. This makes the stock deal a natural placebo: the same withholding bind, minus the cash commitment. A signal that concentrates in cash deals rather than stock deals is therefore informative on its own terms, even though why the two should differ is a question we leave open. The two dimensions also run on different clocks: the anticipatory signal tracks the undisclosed information and resolves once the deal is announced, while the cash commitment serves the purchase itself and—mechanically—persists until it is paid at completion. The contrast is one of timing: the uncertainty is resolved by the announcement, the cash is not.

This paper's nearest neighbors read deal-related language along dimensions other than ours. Thewissen et al. (2024) show that acquirers manage the tone of their earnings press releases: stock bidders inflate disclosure tone by about 15 percent in the year before a stock-for-stock announcement. Ragozzino and Reuer (2024) find a parallel pattern in the spoken record—on the earnings calls of firms active in M&A, the volume of corporate-strategy vocabulary rises, with analysts using such terms nearly 9 percent more, and executives 7.2 percent more, than at inactive firms. Both study language that is deliberately deployed or topic-driven, and both concern tone or subject matter rather than expressed uncertainty. Separately, Keown and Pinkerton (1981) document that prices often move ahead of a deal: roughly half of the market's reaction to a merger occurs before its first public announcement. That anticipatory signal, however, lives in prices, not in language. The cell these literatures leave empty is the one this paper occupies: uncertainty language in the unscripted Q&A, in the anticipatory window before the deal is public, read for what it may reveal about a deal the firm is withholding rather than as language crafted to move the market. To our knowledge, no prior work reads this; we offer it as a positioning claim about where the contribution sits, not as a tested mechanism.

Two readings of this prediction are observationally equivalent, and the framework does not require us to choose between them. Under the first—compliance-constrained disclosure—the executive is unable to confirm the pending deal and barred from denying it, is left to answer around it, and uncertainty rises as a by-product of the bind. Under the second—strategic silence—the

reticence is deliberate: the executive chooses to stay vague about a development that is not yet public. Both would produce the same observable signature: elevated uncertainty language in the Q&A while the deal is undisclosed, resolving at announcement. Our design cannot distinguish them. What we characterize is therefore a correlational pattern keyed to the disclosure state—an anticipatory, cash-concentrated component of CEO uncertainty language that tracks a deal’s passage from private to public—and not a mechanism. We identify no causal channel, establish no intent, and read the relationship descriptively throughout. It is in this deliberately bounded register that the empirical strategy of the following sections puts the pattern to the test.

2.2 Hypothesis Development

The preceding section isolates two dimensions of the pattern we expect. The first is an *anticipatory* dimension, in which CEO uncertainty language is elevated while an acquisition remains undisclosed and recedes once it is announced. The second is a *cash-concentrated* dimension, in which that elevation is stronger for cash acquisitions than for stock. This section turns those dimensions into formal, falsifiable predictions about *where* and *when* the signal appears in a CEO’s unscripted answers, not why it appears, a mechanism the framework leaves open. The object of each prediction is the call-varying residual of CEO uncertainty language in the Q&A, the component that remains once a CEO’s persistent speaking style is netted out (defined formally in the next section), and every prediction is read in descriptive, correlational terms. Timing is measured in event time: e counts calendar quarters relative to a firm’s first acquisition announcement, so that $e = -1$ denotes the call in the quarter immediately before it, and the focal pre-announcement indicator is $\text{PreAnnounceQtr} = \mathbf{1}[e = -1]$, defined for a firm’s first acquisition financed at least half in cash. To state the predictions compactly, we write θ_e for the shift in residual Q&A uncertainty at event position e relative to a firm’s own baseline quarters (for cash acquirers, unless a deal-type superscript such as stock indicates otherwise), and κ_e for the analogous shift in its cash-to-assets ratio; each hypothesis below is a sign restriction on these shifts, which the designs of Section 2.4 estimate. The three hypotheses, H1, H1a, and H1b, are stated both informally and formally and then confronted with the data.

The first hypothesis concerns timing. While the deal is withheld, the executive faces questions that bear on it but can neither address nor deny it, so residual uncertainty should run high in the quarter before the announcement and subside once the deal is public. That this is not mechanical is clear from the nearest evidence, which runs the other way. Thewissen et al. (2024) document that acquirers manage the tone of their disclosure ahead of stock-for-stock deals, a deliberate adjustment; H1 predicts a different register in a different setting, not tone managed to move the market but uncertainty language that surfaces in a cash acquirer’s answers while the deal is withheld. Consistent with the bounded reading established earlier, the hypothesis takes no stance on whether that surfacing is compliance-constrained or strategically chosen; it asserts only that the language moves, and when.

H1 (Timing).

Informally: residual Q&A uncertainty is higher in a cash acquirer’s pre-announcement quarter than in that same firm’s other quarters.

Formally: $\theta_{-1} > 0$.

The second hypothesis sharpens the first by asking where the elevation should concentrate. Cash and stock acquisitions share the disclosure setting, both material and both withheld, yet

differ in one respect the design exploits: a cash purchase draws on an accumulated cash position, whereas a stock exchange need not. As developed in the framework, this makes the stock deal a natural placebo, the same withholding bind without the cash commitment, against which a cash acquirer's run-up can be read. The claim is one of concentration rather than strict specificity, and it concerns the run-up in language alone; H1a makes no claim that the underlying cash accumulation is itself peculiar to cash acquirers, and it asserts no cause.

H1a (Concentration).

Informally: the pre-announcement elevation in residual uncertainty is larger for cash acquirers than for stock acquirers.

Formally: $\theta_{-1}^{\text{cash}} > \theta_{-1}^{\text{stock}}$.

The third hypothesis concerns resolution. The residual uncertainty and the acquirer's cash run on different clocks. The uncertainty tracks the undisclosed information and should resolve at announcement, receding once the deal is public even before it closes, whereas the cash that funds the purchase persists on the balance sheet until it is paid at completion. The most discriminating window is the gap between announcement and completion, when the deal is announced but not yet closed: there the language should already have returned to baseline while the cash still sits unspent. That the cash leg is largely mechanical, since cash leaves only when the deal is paid, is by design; H1b predicts the divergence of the two paths, not the level of either.

H1b (Resolution).

Informally: residual uncertainty returns to baseline as soon as the deal is announced, even before it closes, whereas the cash that funds it remains in place until completion.

Formally: $\theta_{-1} > 0$ and $\theta_{\text{gap}} = 0$, while $\kappa_{\text{gap}} > 0$.

One alternative to this anticipatory reading should be flagged. The pre-announcement rise might reflect not the disclosure bind but analyst scrutiny: analysts spending more of the call on cash and liquidity, mechanically drawing out hedged answers. This is a plausible alternative driver of the run-up, distinct from the framework's two open readings rather than a third version of them, and the design treats it as an identification threat to confront. Its construction and motivation are developed in Section 2.5, and it is put directly to the data as a side analysis in Section 4.1. Flagging it here records that the run-up's most immediate competing explanation is tested, with the verdict reported there; confronting it bounds the alternatives without establishing the disclosure mechanism, which the framework leaves open.

2.3 Estimation of the Main Variable

The dependent variable is built from the language of the call. Following the apparatus adopted in Section 2.1, we measure uncertainty with the Loughran and McDonald finance-specific uncertainty word list, applied as a word share. For a CEO’s answers in the question-and-answer session, the raw measure is the share of uncertainty words among all the words the CEO speaks when answering analysts, which is Equation (2) of Dzielinski et al. (2021):

$$\text{UncAnsCEO} = \frac{\text{UnctWordsAnsCEO}}{\text{WordsAnsCEO}}. \quad (2.1)$$

This raw share is the quantity the decomposition below operates on. The same word-share form is computed for the scripted presentation, UncPreCEO , and for the analysts’ own questions, UncQue ; these enter the decomposition as controls, not as the dependent variable. How the transcripts are tokenized and the words matched is a construction detail we defer to Section 3.1; here the measure is taken as given, and the question is which part of it can carry an anticipatory signal.

An anticipatory, deal-tracking signal must vary from call to call within a single CEO’s tenure, so, as established in Section 2.1, it can reside only in the call-varying part of this measure and not in the CEO’s persistent speaking style. To separate the two, we adopt the decomposition in Equation (4) of Dzielinski et al. (2021), estimated on our sample:

$$\text{UncAnsCEO}_{i,t} = \alpha + \gamma_i \text{CEO}_{i,t} + \beta_s \text{Speech}_{s,i,t} + \beta_k \text{FirmChars}_{k,i,t} + \text{Year}_t + \varepsilon_{i,t}, \quad (2.2)$$

where $\text{CEO}_{i,t}$ is an indicator for manager i being CEO in quarter t , $\text{Speech}_{s,i,t} = \{\text{UncPreCEO}, \text{UncQue}, \text{NegCall}\}$ collects other linguistic features of the same call, $\text{FirmChars}_{k,i,t} = \{\text{SurpDec}, \text{EPSgrowth}, \text{StockRet}, \text{MarketRet}\}$ collects contemporaneous performance, and Year_t absorbs common annual conditions. The persistent component is the CEO fixed effect, summarized as $\text{ClarityCEO}_i = -\gamma_i$, and the object of interest is its residual term, $\text{UncResCEO} = \varepsilon_{i,t}$, which we estimate on our data. Because UncResCEO is taken net of the persistent style and of these observable call-level controls, it is narrower than a style-only residual: a sharper version of the call-varying component located in Section 2.1, not a different object. The authors themselves construct this residual and use it as an explanatory variable in their own market-response model (their Equation (5)); our contribution is not the measure but the anticipatory, disclosure-state reading we give it around undisclosed deals. Finally, because the residual is net of the scripted presentation, UncPreCEO , it isolates the unscripted component, the place where the anticipation of an undisclosed deal would surface; should some of that anticipation also reach the prepared remarks, the residual understates it and so reads as a conservative floor. By the same construction it is net of analyst-question uncertainty, UncQue , and of call negativity, NegCall , a property Section 2.5 relies on when it separates the signal from analyst scrutiny.

One feature of this construction deserves note. Because `UncResCEO` is itself the residual from a first-stage regression and then serves as the dependent variable in the analyses that follow, the design is a standard two-step one, and conventional second-stage standard errors can be understated unless a two-step correction is applied. We flag the property rather than treat it as a defect: it is generic to two-step estimation, Dzielinski et al. themselves use the residual as a regressor, and where it matters a bootstrap that re-estimates both stages supplies a remedy. The deeper question, whether the residual reflects genuine uncertainty rather than merely the analyst scrutiny a call attracts, is not settled by construction; we treat it as an empirical matter and take it up in Section 2.5.

2.4 Methodology and Empirical Design

Each hypothesis of Section 2.2 maps to an estimating equation, and all three share a common panel frame: a within-firm, two-way fixed-effects regression of an outcome on event-time treatment indicators, estimated on quarterly firm observations with standard errors clustered by firm. The main design tests H1. Writing Y_{it} for the outcome of firm i in quarter t ,

$$Y_{it} = \beta \text{PreAnnounceQtr}_{it} + \gamma' X_{it} + \phi_i + \tau_t + \varepsilon_{it}, \quad (2.3)$$

where X_{it} is a vector of firm controls (leverage, log assets, Tobin's Q , return on assets, capital expenditure, a dividend indicator, and cash-flow volatility), ϕ_i is a firm fixed effect, and τ_t is a calendar year-quarter fixed effect. The treatment $\text{PreAnnounceQtr}_{it} = \mathbf{1}[e = -1]$ marks the call in the quarter before the firm's first cash acquisition; post-announcement quarters ($e \geq 0$) are dropped, so the design reads the run-up and not the aftermath, while firms with no cash acquisition, for which e is undefined, anchor the calendar-time effects. The coefficient β is the estimand: it estimates the pre-announcement shift θ_{-1} of Section 2.2, a within-firm contrast between a treated firm's pre-announcement quarter and its own ordinary quarters. We read it descriptively, as an anticipation contrast, and make no causal or parallel-trends claim.

H1b requires tracing the outcome across event time rather than at a single point, so the main design generalizes to an event study that replaces the single indicator with a set of event-time bins:

$$Y_{it} = \sum_b \beta_b \text{Bin}_{b,it} + \gamma' X_{it} + \phi_i + \tau_t + \varepsilon_{it}, \quad (2.4)$$

with four bins: PRE2 ($e = -2$), PRE1 ($e = -1$), GAP (announced but not yet completed), and POST (completed). The omitted baseline is the set of quarters at $e \leq -3$ together with never-acquirers; the post-announcement window is capped at four quarters and truncated at a firm's next announcement. The discriminating window is the gap. If residual uncertainty has resolved by announcement, its coefficient should fall to zero at GAP, when the deal is public but the cash has not yet left, so a Wald test of $\beta_{\text{PRE1}} - \beta_{\text{GAP}} > 0$ carries the empirical content of H1b; the same specification, estimated for the cash ratio in place of uncertainty, traces the persistence of the cash on the other clock, a persistence that is largely mechanical, since the cash is paid only at completion and so sits on the books through the gap. PRE2 is a pre-trend check, expected to be near zero. The bins are estimated for cash acquirers and, as a placebo, for stock acquirers, and every coefficient and Wald contrast is reported two-tailed.

H1a is a claim about the difference between cash and stock acquirers, so it is tested in a single pooled specification carrying both treatments at once:

$$Y_{it} = \beta_c \text{PreAnnCash}_{it} + \beta_s \text{PreAnnStock}_{it} + \gamma' X_{it} + \phi_i + \tau_t + \varepsilon_{it}, \quad (2.5)$$

where PreAnnCash_{it} and PreAnnStock_{it} mark the quarter before a firm’s first cash and first stock acquisition. The sample keeps only the quarters before a firm’s first deal of either type, plus firms that make neither, which form the fixed-effects baseline. The formal cash-specificity test is the single linear restriction $\beta_c - \beta_s > 0$, evaluated by a Wald test; this difference, and not a comparison of one significant coefficient against one insignificant one, is what estimates $\theta_{-1}^{\text{cash}} - \theta_{-1}^{\text{stock}}$ in H1a. When the outcome is the cash ratio rather than residual uncertainty, the specification adds the cash ratio’s own one-quarter lag, so the pre-announcement coefficient tracks the change in cash rather than its level; the residual outcome takes no such lag.

The identifying assumptions are descriptive and common to the three designs. Identification is within firm, through the firm fixed effects, and within calendar time, through the year-quarter fixed effects; the estimand in each case is a within-firm pre-announcement mean shift, measured against a treated firm’s own ordinary quarters; firms that never acquire anchor the calendar-time effects rather than the within-firm contrast itself. The PRE2 bin is a validity device rather than an identification proof: a pre-trend near zero is consistent with the reading, but we claim no parallel-trends result and no causal effect. The mapping from design to hypothesis is one to one: the single-indicator regression tests H1, the event study tests H1b, and the pooled interacted regression tests H1a. The control pool itself differs across designs: in the single-indicator and event-study regressions a firm remains in the baseline until its first cash acquisition, so stock-only acquirers sit among the controls, whereas the pooled regression restricts the baseline to firms that make neither a cash nor a stock acquisition. Two distinct control layers operate and should not be conflated. The residual outcome is already net of a first set of controls, the linguistic and performance variables of the decomposition in Section 2.3, estimated with CEO and year fixed effects; the designs here add a second set, the firm-financial controls X_{it} , with firm and finer year-quarter fixed effects. The two layers serve different purposes and are estimated at different stages.

Several choices that do not belong to a single equation are recorded here. On inference, standard errors are clustered by firm throughout, while some validity tables cluster two ways, by firm and by calendar quarter; focal treatment tests are reported two-tailed. On contamination, deal timing is endogenous, and because only a firm’s first deal sets the event clock, later deals can pollute the baseline; we mitigate this by truncating each firm’s post-window at its second announcement, but the never-acquirer baseline and the first-deal focus remain assumptions, not identification. On functional form, the outcomes are levels or word shares rather than counts, and the cash-ratio regressions carry a partial-adjustment lag because cash is sticky whereas the residual regressions do not. Because that lagged dependent variable is estimated with firm fixed effects, the cash design is a dynamic panel and is subject to the Nickell bias; we therefore re-estimate the cash result without the dynamic term in Section 4.4, where the timing pattern is preserved. Because the residual is a generated regressand from Section 2.3, the two-step standard-error caveat

carries over to every design here. On selection, the residual outcome exists only for CEOs with at least five calls, in the non-financial, non-utility sample where the decomposition is run, so the uncertainty regressions describe a larger, more-covered set of firms than the cash-ratio regressions, a limit on external validity we disclose. On interpretation, the event clock is anchored to the public announcement rather than to a deal's initiation, so reading the pre-announcement quarter as a window in which material information is already being withheld is an interpretive assumption; we observe when a deal is announced, not when it began or became material. The firm-financial controls follow the standard cash-determinants regression, read mechanically, with cash treated as a by-product of retained free cash flow rather than a war chest built for a deal.

2.5 Specification and Measurement of Key Constructs

Before the residual can carry the weight the empirical strategy places on it, its measurement warrants a check. We first confirm that estimating the decomposition we adopt reproduces the published estimates, then ask whether the residual moves with established measures of uncertainty and risk, and, separately, whether the pre-announcement rise might instead reflect the volume of analyst attention to cash rather than the disclosure bind. The first two, construction and convergent validity, are tested in this section; the last is a plausible alternative driver, developed below and put to the data as a side analysis in Section 4.1. By construction the residual is already net of the scripted presentation and of the wording of analysts' questions (Section 2.3); the checks here ask whether it behaves like a measure of uncertainty and whether the scrutiny rival can explain it away. Validity is treated as an empirical question, not an assumption.

We begin with the measure's construction. Because the entire empirical strategy rests on the residual, we first confirm that estimating Equation (4) on our sample reproduces the estimates of Dzielinski et al. (2021). Table 5.15 places the two side by side, variable for variable. The dominant control, the uncertainty of the CEO's own presentation, which absorbs persistent firm and speaking style, loads almost identically, at 0.089 in our baseline against 0.093 in theirs, and the other two speech controls carry the same sign and significance. The fit aligns as well: Dzielinski et al. report the controls' incremental explanatory power over a benchmark of CEO fixed effects alone, 0.054 above a base of 0.31 for a total near 0.36, against a total R^2 of 0.369 here. The samples are not the same, ours covering non-financial, non-utility firms from 2002 to 2018 and theirs from 2003 to 2015, so the firm-performance controls, which we standardize for numerical stability, match in sign rather than in magnitude and our estimation sample is smaller. The point is narrow and sufficient: the residual we carry forward is built by the procedure Dzielinski et al. validate, so the construct validity they establish for the measure transfers to our setting, leaving its behavior in our data as the question for the checks that follow.

The residual is significantly and positively associated, in one-tailed tests, with all three established measures of uncertainty and risk. It rises with firm-level political risk, Hassan et al. (2020)'s scaled measure of the share of a firm's earnings call devoted to political risk (coefficient 0.0001, $p < 0.01$; Table 5.6); with United States economic policy uncertainty, the newspaper-based index of Baker et al. (2016) (0.0124, $p < 0.05$; Table 5.7); and with global economic policy uncertainty, the GDP-weighted index of Davis (2016) (0.0181, $p < 0.05$; Table 5.8). These estimates use industry fixed effects; the positive associations are robust to firm fixed effects, with the United States index then significant at the ten percent level. In economic terms, a one-standard-deviation increase in each measure raises residual uncertainty by about 5 percent of its standard deviation for firm-level political risk, 1.5 percent for United States policy uncertainty, and 2.2 percent for global policy uncertainty.

One channel deserves separate treatment, because it is the most immediate rival to the anticipatory reading. We construct CashScrutiny, the share of analyst question-and-answer turns devoted to cash and liquidity (the word list that identifies these turns is reported in Appendix I), and an indicator HighCashScrutiny for calls above the median. The measure behaves as intended: a firm's cash position predicts how much analysts ask about cash (0.7530 and 0.8519; Table 5.9), so it tracks what it is built to track. It is also a plausible alternative driver of the run-up, because cash scrutiny itself rises in the quarter before a cash acquisition, so it co-moves with both the cash position and the event and could in principle generate the pre-announcement rise mechanically. We therefore treat it as an identification threat and test it directly, as a side analysis whose results are reported in Section 4.1. The measure is the volume of cash questions, not their uncertainty wording: the latter, UncQue, is already netted out inside the residual by the decomposition of Section 2.3, so this test addresses a distinct path that the residual does not already absorb.

The remaining constructs used in the designs are defined here. CashRatio is the firm's cash-to-assets ratio. PreAnnounceQtr, defined in Section 2.2 and used in Section 2.4, marks the quarter before a firm's first cash acquisition. CashScrutiny and HighCashScrutiny are as defined above. The remaining control variables, the firm-financial set of Section 2.4 together with the linguistic and performance controls of the decomposition, are defined where each is introduced. Summary statistics appear in Table 5.1.

Chapter 3

Main Empirical Analyses

3.1 Data, Sample, and Variable Construction

The analysis combines five data sources. Capital IQ supplies the earnings-call transcripts of United States public firms over 2002 to 2018, parsed to speaker role and to the presentation and question-and-answer segments of each call. Quarterly Compustat North America fundamentals supply firm financials, taken from the most recent fiscal quarter ending on or before the call. CRSP daily records and IBES earnings records supply the stock-return, market-return, and earnings-surprise controls that the decomposition of Section 2.3 requires. SDC supplies the merger and acquisition records, including each deal’s announcement date, its completion or withdrawal date, and its cash-versus-stock payment composition. The identity of the chief executive on each call, the input to the manager fixed effect of Section 2.3, comes from Execucomp, whose annual records we aggregate into a monthly panel of which executive holds the CEO role in each quarter. These sources are keyed to different identifier systems, so they must be linked: SDC keys deals by the acquirer’s six-digit CUSIP while the call panel is keyed by Compustat firm identifier (gvkey), and we join deals to calls through the acquirer’s six-digit CUSIP via the CRSP-Compustat link table recorded in the master-sample manifest. The cash-versus-stock classification and the event clock that the designs below depend on exist only through this match, which we record as a researcher degree of freedom rather than assume away.

The uncertainty measures are built in-house from the language of the call. Each transcript is tokenized, and the Loughran and McDonald uncertainty-word share is computed separately by speaker and by segment. The CEO’s question-and-answer share, `UncAnsCEO`, is the input that the first stage of the Section 2.3 decomposition, Equation (4) of Dzielinski et al. (2021) estimated on our sample, residualizes into `UncResCEO`, the call-varying component net of the CEO’s persistent

speaking style. What we discharge here is the construction mechanic only: the equation itself, and the definition and justification of the residual, are set out in Section 2.3 and are not re-derived.

The remaining key estimation variables are constructed and pointed back to their defining homes rather than re-derived here. *UncResCEO* is defined in Section 2.3. $\text{PreAnnounceQtr} = \mathbf{1}[e = -1]$ marks the quarter before a firm's first qualifying deal, as defined in Sections 2.2 and 2.4. *CashRatio* is cash and equivalents over total assets, *cheq/atq*, as defined in Section 2.5 and the Appendix. *CashScrutiny* is the share of analyst question-and-answer turns devoted to cash and liquidity, read from the locked word list in Appendix I, and *HighCashScrutiny* flags any such scrutiny at all, since the median call draws none; both are defined in Section 2.5, and only about eleven percent of calls draw cash scrutiny (the *HighCashScrutiny* mean is 0.1127, Panel A). The firm-financial controls, leverage, log assets, Tobin's Q , return on assets, capital expenditure, a dividend indicator, and cash-flow volatility, are catalogued in Appendix II.

The estimation samples are built in three layers, and it is this layering that makes the number of observations vary from one table to the next. The base call panel comprises every matched Capital IQ call of a United States public, non-financial, non-utility firm over 2002 to 2018, excluding CEOs with fewer than five calls, for whom no speaking style and hence no residual can be estimated. Because the manager fixed effect of Section 2.3 identifies each CEO from Execucomp, the sample is bounded to the firms it covers, approximately the S&P 1500, which skews the uncertainty regressions toward larger, more heavily covered firms. On top of it sits the SDC deal universe: United States public acquirers over the same period with deal status in completed, pending, or withdrawn and a known payment composition, where deals at least half in cash form the cash arm and deals at least half in stock the stock placebo arm, each firm's first qualifying deal setting its event clock. Both payment thresholds are inclusive, so a deal at least half in cash and at least half in stock satisfies both; few do so, 212 deals, well under one percent of either arm, and we impose no tie-break, so a firm whose first cash and first stock acquisitions are one such deal would enter both the cash arm and the stock placebo. The third layer is a set of design-specific trims: the run-up test drops treated firms' post-announcement quarters, the event studies cap the post-window at four quarters and truncate it at the firm's next announcement, and the scrutiny validity and channel tests require at least three analyst question-and-answer turns per call. Each test thus keeps only the firms for which its particular comparison is defined. The five-call requirement enters here as a build mechanic; its consequences for what the residual subsample represents are treated as a selection question in Section 2.4.

Table 5.1 reports summary statistics for the main estimation sample, an anchor of 88,205 earnings-call observations across 1,884 firms over 2002 to 2018. The number of observations varies by row because each variable is summarized on the complete cases of its own anchor panel. The residual *UncResCEO* is centered near zero, with a mean of -0.0000 and a standard deviation of 0.3010 (Panel B); the cash ratio has a mean of 0.1708 (Panel C) and the pre-announcement

indicator a mean of 0.0094 (Panel A). Because the residual exists only on the at-least-five-call, non-financial and non-utility universe, the uncertainty regressions describe a more-covered set of firms, skewed toward larger ones, than the cash-ratio regressions, and for that reason every table that follows reports its own firm and observation counts. We read these statistics descriptively, as the foundation the results sections build on rather than as findings in themselves.

3.2 Main Analysis 1: The Pre-Announcement Run-Up

The first main analysis tests H1 with the single-indicator, two-way fixed-effects regression of Section 2.4, carrying firm and calendar year-quarter fixed effects and standard errors clustered by firm. It is estimated four times, once for each outcome, the cash ratio and the residual `UncResCEO`, in each arm, the cash acquirers (at least half in cash) and the stock acquirers (at least half in stock) that serve as a placebo. Treated firms' post-announcement quarters are dropped, so the design observes only the run-up and not the aftermath. Identification is within firm, setting a treated firm's pre-announcement quarter against its own ordinary quarters; the estimating equation is the one given in Section 2.4 and is not restated. The coefficient on `PreAnnounceQtr` estimates the pre-announcement shift θ_{-1} , and we read it descriptively, as an anticipation contrast, with no causal or parallel-trends claim.

In the cash arm both clocks move in the final pre-announcement quarter. Residual question-and-answer uncertainty is elevated (coefficient 0.0461, $p < 0.01$; column 2 of Table 5.2), and the cash ratio is rising (coefficient 0.0051, $p < 0.01$; column 1). The cash-ratio coefficient tracks the change in cash rather than its level, because that regression carries the cash ratio's own one-quarter lag as a partial-adjustment term (coefficient 0.7990, $p < 0.01$; column 1). The residual elevation is the load-bearing result, and although the table reports the treatment coefficient one-tailed, it survives a two-tailed test as well ($t = 2.68$, $p = 0.0074$), so the result does not rest on the reporting convention.

The two cash-arm columns are not estimated on the same set of firms, because the residual exists only for CEOs with at least five calls. The uncertainty test in column 2 uses 27,622 firm-quarters across 1,248 firms, whereas the cash-ratio test in column 1 uses 67,590 firm-quarters across 2,232 firms. The uncertainty test is therefore the more demanding of the two, run on the smaller and more heavily covered universe, a comparability limit we disclose so the two columns are not read as a single sample.

The effect is material but modest. The uncertainty coefficient equals roughly fifteen percent of a residual standard deviation, about 15.3 percent against the standard deviation of 0.3010 in Panel B of Table 5.1, for a variable from which everything persistent has already been removed, and it appears in a quarter that is itself about one percent of the sample (the mean of `PreAnnounceQtr` is 0.0094, Panel A). The cash coefficient adds roughly half a percentage point of assets to the cash stock in a single quarter, about 3.0 percent of the mean cash ratio of 0.1708 (Panel C). The magnitudes are economically meaningful without being large, which is the register in which we read them.

The stock arm is the placebo. Under the same disclosure constraint, and with an otherwise nearly identical model, neither outcome shows a comparable positive run-up: residual uncertainty

is, if anything, lower (coefficient -0.0429 , not significant; column 6 of Table 5.2) and the cash ratio is flat (coefficient -0.0036 , not significant; column 5). The two models are close to identical in their dynamics, the stock partial-adjustment lag of 0.8013 standing almost exactly where the cash lag of 0.7990 does, so what differs at the pre-announcement quarter is the response and not the specification. A side-by-side reading of this kind is not yet a formal test of the cash-versus-stock difference; that test is supplied in Section 3.4, the third main analysis.

3.3 Main Analysis 2: Differential Timing Around the Announcement

The second main analysis replaces the single pre-announcement indicator with the four-bin event study of Section 2.4. The bins are PRE2 at two quarters before announcement, PRE1 at one quarter before, GAP for the interval in which the deal is announced but not yet closed, and POST for the quarters after completion. The post-completion bin is therefore formed only from deals that close, since the quarters following a withdrawal are dropped from the event clock rather than carried into it. The baseline is formed by quarters three or more before announcement together with never-acquirers. The specification is otherwise that of the first main analysis, and because no ordering is imposed across the bins every coefficient is reported two-tailed. The test is run on the matched universe in which both the residual and the cash ratio are defined, 28,102 firm-quarters across 1,320 firms, so that both outcomes are estimated on the identical rows; estimating them on the same rows is what lets the two paths be read against each other while guarding the differential-timing reading against a sample-composition artifact. The estimating equation is the one given in Section 2.4 and is not restated; the reading remains within firm and correlational, with no identification claim.

The pre-trend is flat on both clocks. Two quarters before announcement, at PRE2, the coefficient is null for the residual (coefficient 0.0068, not significant; column 1 of Table 5.3) and for the cash ratio (coefficient 0.0008, not significant; column 2), so the one-quarter-before estimates are not the tail of a pre-existing trend. This is a validity device, not a proof of identification.

The discriminating contrast is the drop at the gap. On the information clock, residual uncertainty spikes one quarter before announcement (coefficient 0.0473, $p < 0.01$; column 1 of Table 5.3) and falls to insignificance the instant the deal is public (coefficient at GAP of 0.0018, not significant), a significant decline from the pre-announcement peak to the gap (drop 0.0455, $p < 0.05$) and a large significant swing from the peak to completion (drop 0.0723, $p < 0.01$). On the transaction clock, the cash ratio is elevated one quarter before announcement (coefficient 0.0061, $p < 0.05$; column 2), does not fall when the deal is announced (the pre-announcement-to-gap drop of 0.0006 is not significant), and declines only once the cash leaves at completion (drop from gap to completion 0.0210, $p < 0.01$; the completion coefficient itself is -0.0155 , $p < 0.01$). Because the gap-quarter cash coefficient itself, 0.0055, is positive but not significant, the persistence of cash rests on the absence of a decline at announcement rather than on a significant gap-quarter level. The contrast between these two paths is the claim, read within firm and correlationally.

The pre-announcement spike is economically the same event as the one the first main analysis measured: 0.0473 here against 0.0461 there, about fifteen percent of a residual standard deviation, 15.7 percent against the 0.3010 in Panel B of Table 5.1. What the event study adds is the round

trip. The swing from the pre-announcement peak to completion is a drop of 0.0723 ($p < 0.01$). The uncertainty raised before announcement has already fallen to a level indistinguishable from zero once the deal becomes public, at the gap; the remainder of the swing is the further move below baseline after completion. The cash leg, for its part, is identified off the same well-behaved partial-adjustment specification used in the first main analysis, its one-quarter lag standing at 0.7547 ($p < 0.01$; column 2 of Table 5.3). Because this event study runs on the matched universe rather than the all-firm sample, scaling by the all-universe standard deviation is a disclosed approximation, and the magnitude is read as supportive rather than decisive.

The same round trip appears in the cash arm of the placebo event study, estimated on its own complete-case cash universe of 29,535 firm-quarters across 1,326 firms. There residual uncertainty again spikes one quarter before announcement (coefficient 0.0486, $p < 0.01$; column 1 of Table 5.4) and collapses to insignificance at the gap (coefficient 0.0058, not significant), with a significant decline from the peak to the gap (drop 0.0428, $p < 0.10$) and a significant swing from the peak to completion (drop 0.0681, $p < 0.01$). This corroborates the matched-universe result on the placebo cash sample, which largely overlaps it rather than forming an independent one. Only the cash column of this placebo table is used here; its cash-versus-stock contrast is reserved for Section 3.4.

The claim rests on the contrast between the two paths, not on the magnitude or the sign of any single drop. Uncertainty tracks the information and reverses at announcement; cash tracks the transaction and falls at completion, a persistence that is in part mechanical, since the cash leaves the balance sheet only at closing. We do not over-read the below-baseline completion coefficient on the uncertainty clock (coefficient -0.0250 , $p < 0.10$; column 1 of Table 5.3); it is flagged rather than interpreted. Because the post-completion bin is built only from deals that close, the contrasts that reach it could in part reflect which deals complete; the announcement-quarter contrast does not, since the gap is dated to the public announcement, and Section 4.3 re-estimates the event study with a post bin that counts a withdrawal as a resolution, where the decline is essentially unchanged. The reading throughout is correlational and descriptive, supportive of the disclosure-state pattern rather than definitive about any mechanism, which we leave open.

3.4 Main Analysis 3: Cash-Specificity

A first step reruns the event study side by side on the stock placebo arm, using the stock column of Table 5.4 that was held back from the previous analysis. Neither arm trends into the event: the pre-trend coefficient two quarters out is small and insignificant in both, 0.0105 for the cash arm and -0.0056 for the stock arm. From there the two diverge. The cash arm reproduces the run-up, with a spike one quarter before announcement (coefficient 0.0486, $p < 0.01$) and a swing from the peak to completion (drop 0.0681, $p < 0.01$), while the stock arm shows none of it (coefficient one quarter before announcement of -0.0404 , not significant) and, if anything, moves the other way across announcement (decline from the pre-announcement quarter to the gap of -0.0756 , $p < 0.10$). This is a descriptive comparison; it motivates a formal test rather than standing in for one.

A side-by-side display of this kind implies a comparison it never actually runs. The formal test instead uses the pooled, interacted model of Section 2.4, which carries both treatments at once and tests the cash-minus-stock difference directly through a single Wald linear restriction, $\beta_c - \beta_s > 0$. That restriction is what estimates $\theta_{-1}^{\text{cash}} - \theta_{-1}^{\text{stock}}$ in H1a, and it is this difference, not a comparison of one significant coefficient against one insignificant one, that constitutes the cash-specificity test. The pooled baseline is the set of firms that make neither a cash nor a stock acquisition. The reading remains correlational, with no identification claim, and the framing is one of concentration rather than strict specificity.

On the effect, the pooled model delivers the formal cash-specificity result. The cash arm's pre-announcement uncertainty coefficient is significant (coefficient 0.0459, $p < 0.05$; column 1 of Table 5.5) and the stock arm's is insignificant and negative (coefficient -0.0524 , not significant). The formal cash-minus-stock difference, evaluated as the single Wald restriction, is 0.0983 (standard error 0.0476, $z \approx 2.07$, $p = 0.039$), significant at the five percent level on a two-tailed test. Because the two arms move in opposite directions, the difference is larger than either coefficient, roughly a third of a residual standard deviation, about 32.7 percent against the 0.3010 in Panel B of Table 5.1. It is this formal linear restriction, and not the contrast between a significant and an insignificant coefficient, that carries the cash-specificity claim.

On the proposed cause, the picture is weaker. The cash build-up itself is only faintly cash-sided: the cash-minus-stock difference in the cash ratio is insignificant on the matched universe (difference 0.0064, not significant; column 2 of Table 5.5) and reaches only marginal significance on the full panel (difference 0.0092, $p < 0.10$; column 3). The effect concentrates in cash deals, but the war-chest mechanism that might explain it is therefore not established, and the mechanism is left open.

We therefore read the result as supported but fragile. One formal test at the five percent level

supports H1a's claim of concentration in cash, but its significance rides on the imprecise, negative stock estimate (coefficient -0.0524 , standard error 0.0436 ; column 1 of Table 5.5), and the cause's own cash-specificity is weaker still, marginal and only on the full panel. The wording stays with concentration rather than strict specificity, the difference between a significant and an insignificant coefficient is not treated as the test, and no cause is asserted.

Chapter 4

Additional Analyses

4.1 Ruling Out Analyst Scrutiny

The most immediate rival to the anticipatory reading is analyst scrutiny: that the quarters before a deal draw harder analyst questions about cash, and that the hard questioning, rather than the deal itself, produces the hedged answers. This is a genuine confound and not a strawman. The incidence of cash scrutiny itself rises in the quarter before a cash acquisition (the indicator *HighCashScrutiny* carries a coefficient of 0.0408, $p < 0.05$; column 4 of Table 5.2), so it co-moves with both the cash position and the event and could in principle generate the run-up mechanically. The continuous volume of cash questions rises too but not significantly (coefficient 0.0854, not significant; column 3), so what rises reliably before a cash deal is the incidence of cash scrutiny, not its volume. We address this rival in three steps.

The first step is to confirm that the scrutiny measure is real. The share of analyst question-and-answer turns devoted to cash and liquidity rises strongly with the firm's cash ratio (coefficient 0.7530, $p < 0.01$, in a one-tailed test, rising to 0.8519, $p < 0.01$, with the full control set; columns 1 and 2 of Table 5.9) and with a high-cash indicator (coefficient 0.1754, $p < 0.01$; column 3), so it tracks what it is built to track and the nulls that follow are not the artifact of a dead measure.

The second step asks whether scrutiny alone moves the residual, and it does not. Its direct effect on *UncResCEO* is zero to four decimal places (coefficient -0.0000 , not significant; standard error 0.0013; 41,512 calls; Panel A of Table 5.10), and the null holds under a logit on the dichotomized residual (log-odds coefficient -0.0003 , not significant; Panel B) and under the raw uncertainty measure and coarser scrutiny proxies alike. This test is not redundant with the way the residual is

built: CashScrutiny is the volume of cash questions, whereas the uncertainty wording of those questions, UncQue, is already netted out inside the residual by the decomposition of Section 2.3, so the null speaks to a path the residual does not already absorb.

The third and operative step puts scrutiny and the pre-announcement indicator in the same model, estimated over the cash acquirers' pre-announcement and earlier quarters. The pre-announcement indicator stays significant, both with scrutiny entered (coefficient 0.0413, $p < 0.01$; column 1 of Table 5.11) and with the scrutiny-by-pre-announcement interaction added (coefficient 0.0439, $p < 0.05$; column 2), while scrutiny's own effect remains null (0.0000 and 0.0001, neither significant) and the interaction is insignificant (coefficient -0.0056 , not significant; column 2). The reason raises residual uncertainty; the questioning does not, and the questioning does not amplify the reason.

The register here should stay honest. This is a failure to find, paired with a within-model dissociation, and not a powered test of equivalence. The interaction's confidence interval, approximately $[-0.027, 0.016]$, does not exclude effects that would matter against a run-up near 0.04; about eighty-nine percent of calls draw no cash scrutiny at all; and the contrast between a significant and an insignificant coefficient is not itself a test. The claim is that scrutiny does not account for this particular run-up, not that scrutiny never matters anywhere.

4.2 Outsider Reactions: The Bid-Ask Spread

A natural further question is whether the speech-uncertainty signal travels beyond the firm to the outside information environment. We measure that environment by the bid-ask spread, a standard adverse-selection proxy, taken over the twenty-five trading days that begin on the call date, the post-call window of Bushee et al. (2018), who study the same two-segment structure. Two priors frame the test. Dzielinski et al. (2021) report that neither the presentation nor the residual component of uncertainty is associated with stock price or volume responses, and that the residual explains little of the market reaction, but they do not examine any liquidity outcome. Bushee et al. (2018) show that the language of a call relates to information asymmetry with opposite signs across the call's two segments: complexity in the scripted presentation is positively associated with information asymmetry, while complexity in the spontaneous response is negatively associated with it. We therefore regress the post-call spread on the three speech components of Section 2.3, retaining the lagged spread and the usual controls, and read the components one at a time.

The residual carries no such signal. The coefficient on *UncResCEO* is small and statistically insignificant in every one of the twelve specifications, whether the spread is measured in the call quarter (the six contemporaneous coefficients range from -0.0594 to -0.1021 ; 42,625 calls) or one quarter ahead (43,049 calls); see Table 5.12. Dzielinski et al. (2021) document this market inertness of the residual for price and volume; here it extends to the bid-ask spread, a liquidity outcome they do not test, and it accords with the Bushee et al. (2018) finding that the spontaneous response segment does not widen information asymmetry. The reading is the cautious one: this is an absence of association, not a powered test of equivalence.

The scripted presentation behaves differently. The coefficient on *UncPreCEO* is positive and significant in four of the six contemporaneous specifications under a one-tailed test (0.1664 , $p < 0.05$, column 1; 0.3496 , $p < 0.01$, column 2; 0.2945 , $p < 0.01$, column 4; 0.1644 , $p < 0.10$, column 6), including all three specifications with firm fixed effects; it is insignificant in the two industry-fixed-effects specifications that add the extended controls (columns 3 and 5) and in all six one-quarter-ahead specifications, while *ClarityCEO* is insignificant throughout. This pattern matches the Bushee et al. (2018) result that complexity in the scripted presentation is positively associated with information asymmetry. It does not contradict the Dzielinski et al. (2021) finding of no presentation effect on price or volume, because the bid-ask spread captures information asymmetry rather than a price or volume reaction. The association is, however, contemporaneous and modest, and our linguistic measure is the frequency of uncertainty words rather than the complexity measure of Bushee et al. (2018), so the alignment is one of direction.

Taken together, the two segments of the same call relate to the outside information environment in different ways: the scripted presentation is positively associated with the contemporaneous

spread, while the residual shows no detectable association. We do not test the difference between the two components directly, so we read this as a pattern consistent with a channel difference rather than a tested contrast. This pattern sits within the scripted-versus-spontaneous structure of Dzielinski et al. (2021) and the presentation-versus-response split of Bushee et al. (2018), and both component readings run with the grain of that prior work: the residual's inertness to the spread extends the market inertness Dzielinski et al. (2021) document for price and volume to a liquidity outcome they do not test, while the presentation's positive association aligns in direction with Bushee et al. (2018), who find that greater complexity in the scripted presentation is associated with wider information asymmetry.

4.3 Robustness: Withdrawal as a Resolution Event

The differential-timing result of Section 3.3 dates the post bin to completion: a deal's quarters enter the post bin only once it has closed, and the quarters that follow a withdrawal are dropped from the event clock. Two of the three contrasts that trace the round trip, the decline from the pre-announcement peak to completion and from the gap to completion, are therefore estimated only over deals that complete. Because completing and withdrawn deals need not be alike, part of the post-completion decline could in principle reflect which deals complete rather than the resolution of the disclosure state. The third contrast is not exposed to this: the decline from the pre-announcement peak to the gap is measured at the public announcement, a point every announced deal reaches whatever its eventual outcome, so it conditions on nothing that follows.

We can probe the completion-dated legs directly by letting a withdrawal count as a resolution of the deal in the same way a completion does. A withdrawal ends the disclosure state as surely as a completion: once a pending deal is called off and disclosed, the executive is no longer answering around it. We therefore redefine the post bin so that the quarters at and after a withdrawal enter it alongside the completed quarters, and we re-estimate the event study of Section 2.4 on this resolution-inclusive sample; Table 5.13 reports it. Folding in the withdrawn deals asks whether the post-resolution decline is particular to the deals that complete or appears whenever the disclosure state resolves.

The decline holds. Residual uncertainty still falls from its pre-announcement peak to the resolved quarters, by 0.0687 ($p < 0.01$), against 0.0723 in the main specification, and the announcement-quarter leg, the drop from the peak to the gap, is 0.0457 ($p < 0.05$), almost exactly the 0.0455 of the main result. The cash leg declines from its pre-announcement level to resolution by 0.0204 ($p < 0.01$). Treating a withdrawn deal as a resolution event, rather than discarding it, leaves the round trip in place.

This check is a narrow one, and its main reassurance is structural rather than statistical. Few first deals are withdrawn and then observed for complete-case quarters within the post window, so the resolution-inclusive sample adds only 89 firm-quarters and a single firm to the 28,102 of the main analysis; with so little new data the post-completion decline can barely move, so the result shows that the handful of withdrawn deals behave like the completed ones rather than ruling out a completion-driven reading on its own. The firmer ground is the announcement-quarter leg itself: the decline from the pre-announcement peak to the gap is measured when the deal becomes public and conditions on nothing that follows, so the resolution of the residual at announcement never depended on which deals complete. We read the resolution-inclusive estimates as consistent with the main specification rather than as an independent test of it.

4.4 Robustness: The Cash Result Without the Dynamic Term

The cash-ratio regressions of Sections 3.3 and 3.4 include the cash ratio's own one-quarter lag alongside firm fixed effects, a partial-adjustment form that suits a sticky stock like cash. A dynamic panel of this kind, a lagged dependent variable estimated with fixed effects, is subject to the Nickell bias, which the within transformation induces and which is not confined to the coefficient on the lag alone. Rather than reason about the size of that bias, we guard against it directly, by removing the dynamic term and re-estimating without it. The residual outcome carries no lag and is unaffected, so it serves here as an anchor.

We re-estimate the cash column with static fixed effects, dropping the lagged dependent variable; Table 5.14 reports it. Removing the lag changes what the cash coefficients describe. With the lag, each coefficient is a change in the cash ratio around its own recent level, a partial adjustment; without it, the coefficients describe the level of the cash ratio relative to the baseline. The two specifications are therefore on different scales, and the comparison that carries over is the timing pattern, the sign and significance of the bins across event time, not the size of any single coefficient.

The timing pattern is the same. Without the lag the cash ratio does not fall when the deal is announced, the pre-announcement-to-gap change of -0.0012 being insignificant just as the partial-adjustment change of 0.0006 was, and it falls only once the cash leaves at completion, by 0.0318 from the gap to completion ($p < 0.01$), with a completion coefficient of -0.0189 ($p < 0.01$). The pre-announcement-to-completion decline is 0.0305 ($p < 0.01$), larger than the 0.0216 of the partial-adjustment specification, as expected when the coefficients describe a level rather than a change. The residual column is unchanged, as it must be, since it never carried the lag.

The reading is therefore unchanged. The lag belongs in the main specification, because cash is sticky and the partial-adjustment form is the right description of how it moves; the static specification is not a better model but a check that removes the dynamic term entirely. Because the static specification carries no lagged dependent variable, it carries no dynamic-panel bias, and the cash timing conclusion, no fall at announcement and a fall at completion, holds whether or not that term is present. The timing of the cash path, on which the differential-timing result rests, does not ride on the dynamic specification.

Chapter 5

Conclusion

In this study we examine whether the language a chief executive uses in the unscripted question-and-answer session of an earnings call tracks the disclosure state of an acquisition the firm is still withholding, using a residual measure of chief-executive question-and-answer uncertainty that nets out each executive's persistent speaking style.

We find that residual uncertainty is elevated in the quarter before a cash acquisition, with no comparable rise before stock acquisitions; that it becomes indistinguishable from zero once the deal is announced, even before it closes, while the cash that funds it persists to completion; and that the elevation concentrates in cash acquisitions rather than stock on a formal pooled test.

The pattern is not accounted for by analysts devoting more of the call to cash questions; the pre-announcement elevation survives a direct measure of analyst cash scrutiny and its interaction with the pre-announcement window. A further analysis finds the residual unrelated to the firm's post-call bid-ask spread, while the scripted presentation is positively associated with it, consistent with the scripted and unscripted segments relating to the outside information environment differently. The differential-timing result holds when a withdrawal is treated as a resolution of the deal and when the cash specification is estimated without its dynamic term.

Taken together, these patterns suggest that the unscripted language of earnings calls carries a readable, anticipatory trace of a deal's passage from private to public, a signature investors and researchers can in principle observe before the announcement. For the academic literature, the result speaks to two strands usually studied apart: the theory of when an informed manager withholds private information, and the evidence that markets anticipate corporate transactions before they are announced. The within-firm language signature we characterize sits in the window between them. We draw no prescription for investors, managers, or regulators from a correlational

pattern that identifies no causal channel; the contribution is to characterize a regularity, not to recommend acting on it.

These findings carry clear limitations. The evidence is correlational and within-firm, the design supports no causal identification, and we do not establish why the signal concentrates in cash deals; the war-chest mechanism in particular is unestablished, so two readings, a compliance-constrained inability to speak and a strategically chosen reticence, remain observationally equivalent.

Two further boundaries qualify the evidence. The residual is estimated only for chief executives with enough calls to fix a speaking style, which skews the language sample toward larger, more heavily covered firms, and the scrutiny rule-out is a failure to find rather than a powered test of equivalence, since most calls draw no cash scrutiny at all. The measure itself bounds the inference further. Uncertainty is captured by applying a finance-specific word list to the transcripts, a count of words that abstracts from their context, and the residual is one operationalization of call-specific uncertainty rather than a direct reading of what a chief executive knows. The stock-financed placebo shares the disclosure bind but is an imperfect counterfactual, and the sample, United States public firms from 2002 to 2018, approximately the S&P 1500, may not extend to other periods, other markets, or the smaller firms outside it.

Future work could ask whether the same anticipatory language signature appears around other classes of withheld material events, and could pursue designs able to separate compliance-constrained from strategic silence, which our setting cannot distinguish. The same residual-uncertainty reading could be applied to other corporate transactions, such as divestitures, joint ventures, and strategic alliances, and to settings outside the United States; richer measures of spoken uncertainty could replace the word-list count; and establishing why the signal concentrates in cash acquisitions, the cash-accumulation channel this study leaves open, would require a design the present setting does not provide.

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Table 5.1: Summary Statistics

Variable	Unit	N	Mean	SD	Min	P25	Median	P75	Max
<i>A. Independent Variables</i>									
CashScrutiny	C	76,570	0.4250	1.7576	0.0000	0.0000	0.0000	0.0000	100.0000
GEPU_log	C	88,205	4.6872	0.3672	3.8626	4.3797	4.7238	4.9666	5.5106
HighCashScrutiny	C	76,570	0.1127	0.3162	0.0000	0.0000	0.0000	0.0000	1.0000
PRisk	C	85,995	99.6187	146.5189	0.0000	17.0778	53.3729	120.1973	1192.4308
PreAnnounceQtr	C	88,205	0.0094	0.0963	0.0000	0.0000	0.0000	0.0000	1.0000
US_EPU_log	C	88,205	4.7123	0.3661	3.8018	4.4805	4.6891	4.9625	5.6478
<i>B. Dependent Variables</i>									
BGTLevel_Spread	C	87,018	0.0017	0.0026	0.0001	0.0004	0.0009	0.0017	0.0164
UncAnsCEO	C	71,748	0.7806	0.3925	0.0000	0.5147	0.7415	1.0018	2.5874
UncPreCEO	C	71,727	0.6602	0.3917	0.0000	0.3810	0.6001	0.8723	2.2432
UncResCEO	C	44,900	-0.0000	0.3010	-1.5108	-0.1846	-0.0180	0.1650	1.7794
<i>C. Firm Controls</i>									
Capex	C	87,627	0.0487	0.0536	0.0000	0.0176	0.0324	0.0592	0.5203
CashRatio	C	87,990	0.1708	0.1806	0.0000	0.0368	0.1047	0.2435	0.9938
CashRatio_lag	C	79,131	0.1709	0.1804	0.0000	0.0367	0.1048	0.2443	0.9938
DivDummy	C	88,134	0.4755	0.4994	0.0000	0.0000	0.0000	1.0000	1.0000
EarnVol	C	87,440	0.0701	0.1907	0.0006	0.0163	0.0315	0.0695	19.5101
FirmMat	C	87,571	-0.0432	2.1826	-317.5714	-0.0078	0.2191	0.4195	0.9252
HighCash	C	87,990	0.3333	0.4714	0.0000	0.0000	0.0000	1.0000	1.0000
Leverage	C	87,994	0.2330	0.2240	0.0000	0.0470	0.2065	0.3455	3.9453
NegCall	C	86,858	0.9210	0.3024	0.0000	0.7038	0.8798	1.0943	2.3901
ROA	C	87,552	0.0386	0.1330	-3.9826	0.0146	0.0525	0.0925	0.5490
TobinsQ	C	87,363	2.1363	1.5197	0.3847	1.2681	1.6791	2.4362	35.6144
UncQue	C	81,031	1.4398	0.4831	0.0000	1.1216	1.4108	1.7268	3.5945
lnAssets	C	87,994	7.4710	1.6607	-0.5692	6.3008	7.3763	8.5436	12.4592
sCFO	C	83,662	0.0619	0.2233	0.0014	0.0236	0.0395	0.0664	32.4035

Notes: Summary statistics for the main estimation sample (United States public, non-financial, non-utility firms, 2002–2018). Each variable is summarized on the complete cases of its own estimation universe, so the number of observations varies by row; never-acquirer firm-quarters are included as the fixed-effects baseline and treated firms' post-announcement quarters are excluded. All variables are defined in the main text and in Appendices I and II.

Table 5.2: Pre-Announcement Run-Up Test: Cash versus Stock Acquirers

	Cash acquirers				Stock acquirers (placebo)			
	(1) CashRatio	(2) UncResCEO	(3) CashScrutiny	(4) HighCashScrutiny	(5) CashRatio	(6) UncResCEO	(7) CashScrutiny	(8) HighCashScrutiny
PreAnnounceQtr	0.0051*** (0.0017)	0.0461*** (0.0172)	0.0854 (0.0749)	0.0408** (0.0194)	-0.0036 (0.0035)	-0.0429 (0.0307)	-0.3222 (0.2024)	-0.0303 (0.0313)
CashRatio _{t-1} (partial adj.)	0.7990*** (0.0070)	—	—	—	0.8013*** (0.0057)	—	—	—
Leverage	-0.0146*** (0.0039)	-0.0338 (0.0217)	-0.1521 (0.1470)	-0.0179 (0.0308)	-0.0213*** (0.0034)	-0.0292 (0.0196)	-0.1056 (0.1358)	-0.0428 (0.0262)
lnAssets	-0.0072*** (0.0011)	-0.0037 (0.0082)	-0.0391 (0.0482)	0.0190** (0.0096)	-0.0079*** (0.0008)	-0.0043 (0.0063)	-0.0733** (0.0369)	0.0077 (0.0078)
TobinsQ	0.0013** (0.0006)	-0.0031 (0.0030)	-0.0208 (0.0144)	0.0006 (0.0027)	0.0015*** (0.0004)	-0.0017 (0.0025)	-0.0113 (0.0123)	0.0030 (0.0024)
ROA	0.0053 (0.0042)	0.0499 (0.0321)	-0.5659** (0.2254)	-0.1006** (0.0405)	0.0075* (0.0041)	0.0530* (0.0296)	-0.5350*** (0.1982)	-0.0869** (0.0366)
Capex	-0.1057*** (0.0126)	0.0842 (0.0986)	-1.2215*** (0.3534)	-0.3646*** (0.0880)	-0.1067*** (0.0113)	0.1256 (0.1006)	-1.4479*** (0.3000)	-0.4470*** (0.0793)
DivDummy	-0.0011 (0.0009)	-0.0030 (0.0106)	-0.0561 (0.0537)	-0.0111 (0.0107)	-0.0017** (0.0008)	-0.0137* (0.0080)	-0.0564 (0.0384)	-0.0146* (0.0087)
sCFO	0.0008 (0.0006)	0.0052 (0.0085)	0.1180*** (0.0140)	0.0235*** (0.0028)	0.0007 (0.0007)	-0.0009 (0.0110)	0.0906*** (0.0337)	0.0169*** (0.0057)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cal. Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firms	2,232	1,248	1,237	1,237	2,281	1,338	1,325	1,325
N (firm-quarters)	67,590	27,622	26,216	26,216	84,979	39,377	37,065	37,065
R ²	0.665	0.001	0.002	0.002	0.672	0.001	0.002	0.002

Notes: Pre-announcement run-up test. PreAnnounceQtr marks the quarter before a firm's first qualifying acquisition; post-announcement quarters are dropped and never-acquirers form the baseline. The cash arm is acquisitions at least half in cash; the stock arm is the at-least-half-in-stock placebo. Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Table 5.3: Uncertainty and Cash Around the Announcement (Matched Sample)

	(1) UncResCEO	(2) CashRatio
PRE2 ($t-2$, pre-trend)	0.0068 (0.0178)	0.0008 (0.0024)
PRE1 ($t-1$, pre-announce)	0.0473 *** (0.0178)	0.0061 ** (0.0024)
GAP (announced, pre-close)	0.0018 (0.0187)	0.0055 (0.0034)
POST (completed)	-0.0250 * (0.0130)	-0.0155 *** (0.0022)
Drop: PRE1 – GAP	0.0455 ** (0.0225)	0.0006 (0.0039)
Drop: GAP – POST	0.0268 (0.0199)	0.0210 *** (0.0042)
Drop: PRE1 – POST	0.0723 *** (0.0190)	0.0216 *** (0.0031)
<i>Controls</i>		
Leverage	-0.0415 * (0.0213)	-0.0215 *** (0.0054)
lnAssets	-0.0060 (0.0084)	-0.0074 *** (0.0020)
TobinsQ	-0.0045 (0.0029)	0.0013 ** (0.0006)
ROA	0.0572 * (0.0325)	0.0067 (0.0061)
Capex	0.1073 (0.1054)	-0.1434 *** (0.0141)
DivDummy	-0.0099 (0.0104)	-0.0016 (0.0014)
sCFO	0.0096 ** (0.0043)	-0.0011 (0.0007)
CashRatio $_{t-1}$ (partial adj.)	—	0.7547 *** (0.0108)
Firm FE / Year-Qtr FE / Controls	Yes	Yes
N (firm-quarters)	28,102	28,102
Firms	1,320	1,320

Notes: Disclosure-window event study on the matched universe, where UncResCEO and CashRatio are defined on the identical firm-quarters. Bins: PRE2 ($e=-2$), PRE1 ($e=-1$), GAP (announced, not yet closed), POST (completed); the omitted baseline is $e \leq -3$ with never-acquirers. Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Table 5.4: Event Study by Payment Type: Cash versus Stock Acquirers

	DV: UncResCEO	
	(1) Cash acquirers	(2) Stock acquirers
PRE2 ($t-2$, pre-trend)	0.0105 (0.0174)	-0.0056 (0.0368)
PRE1 ($t-1$, pre-announce)	0.0486*** (0.0174)	-0.0404 (0.0299)
GAP (announced, pre-close)	0.0058 (0.0181)	0.0353 (0.0318)
POST (completed)	-0.0195 (0.0127)	-0.0048 (0.0246)
Drop: PRE1 – GAP	0.0428* (0.0221)	-0.0756* (0.0405)
Drop: GAP – POST	0.0253 (0.0194)	0.0401 (0.0382)
Drop: PRE1 – POST	0.0681*** (0.0187)	-0.0355 (0.0358)
<i>Controls</i>		
Leverage	-0.0372* (0.0211)	-0.0286 (0.0194)
lnAssets	-0.0063 (0.0080)	-0.0041 (0.0062)
TobinsQ	-0.0034 (0.0029)	-0.0018 (0.0025)
ROA	0.0534* (0.0315)	0.0543* (0.0293)
Capex	0.0816 (0.0993)	0.1297 (0.0987)
DivDummy	-0.0065 (0.0100)	-0.0132* (0.0080)
sCFO	0.0049 (0.0084)	-0.0008 (0.0110)
Firm FE / Year-Qtr FE / Controls	Yes	Yes
N (firm-quarters)	29,535	39,819
Firms	1,326	1,363

Notes: Disclosure-window event study estimated separately for cash acquirers and, as a placebo, stock acquirers, each on its own complete-case universe. Bins and baseline as in the matched event study. Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Table 5.5: Cash-Specificity Test: Cash versus Stock Acquirers

	EFFECT	PROPOSED CAUSE	
	(1) UncResCEO (matched)	(2) CashRatio (matched, +lag)	(3) CashRatio (full panel, +lag)
Pre-announce qtr, Cash acquirer	0.0459** (0.0185)	0.0056** (0.0025)	0.0049*** (0.0019)
Pre-announce qtr, Stock acquirer	-0.0524 (0.0436)	-0.0008 (0.0066)	-0.0043 (0.0051)
Cash – Stock (formal test)	0.0983** (0.0476)	0.0064 (0.0071)	0.0092* (0.0055)
<i>Controls</i>			
Leverage	-0.0411* (0.0230)	-0.0236*** (0.0056)	-0.0194*** (0.0043)
InAssets	-0.0034 (0.0092)	-0.0033 (0.0021)	-0.0065*** (0.0011)
TobinsQ	-0.0025 (0.0031)	0.0007 (0.0006)	0.0012** (0.0005)
ROA	0.0556 (0.0355)	0.0046 (0.0065)	0.0054 (0.0046)
Capex	0.0487 (0.1065)	-0.1437*** (0.0148)	-0.1095*** (0.0139)
DivDummy	-0.0068 (0.0109)	-0.0026* (0.0014)	-0.0015 (0.0010)
sCFO	0.0040 (0.0088)	-0.0008 (0.0007)	0.0009 (0.0006)
CashRatio _{t-1} (partial adj.)	—	0.7778*** (0.0113)	0.7957*** (0.0073)
Firm FE / Year-Qtr FE / Controls	Yes	Yes	Yes
UncResCEO-present restriction	Yes	Yes	No
N (firm-quarters)	25,600	24,347	61,876
Firms	1,191	1,176	2,141

Notes: Pooled cash-specificity test: both treatments enter one regression and the cash-minus-stock difference is evaluated by a Wald test; the baseline is firms making neither a cash nor a stock acquisition. Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Table 5.6: Convergent Validity: Firm Political Risk

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
PRisk	0.0001 *** (0.0000)	0.0003 *** (0.0000)	0.0001 *** (0.0000)	0.0002 *** (0.0000)
UncQue	-0.0022 (0.0032)	0.0204 *** (0.0045)	-0.0014 (0.0037)	0.0081 *** (0.0029)
NegCall	-0.0025 (0.0051)	0.1874 *** (0.0141)	-0.0036 (0.0073)	0.1477 *** (0.0082)
lnAssets	-0.0002 (0.0004)	-0.0296 *** (0.0039)	-0.0054 (0.0058)	0.0048 (0.0078)
TobinsQ	-0.0005 (0.0009)	-0.0007 (0.0036)	-0.0008 (0.0022)	0.0001 (0.0023)
ROA	0.0291 ** (0.0145)	0.0477 (0.0318)	0.0540 ** (0.0264)	0.0346 (0.0230)
CashRatio	0.0019 (0.0077)	-0.0354 (0.0357)	0.0206 (0.0240)	0.0265 (0.0287)
DivDummy	-0.0031 (0.0021)	-0.0139 (0.0118)	-0.0108 (0.0074)	-0.0229 ** (0.0103)
FirmMat	0.0012 (0.0013)	0.0021 ** (0.0010)	0.0027 (0.0041)	0.0006 (0.0007)
EarnVol	0.0078 (0.0077)	-0.0072 (0.0157)	0.0056 (0.0117)	-0.0104 (0.0132)
UncPreCEO	-0.0051 (0.0034)		-0.0054 (0.0054)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	44,497	65,760	44,497	65,760
R ²	0.003	0.043	0.003	0.023
Adj. R ²	0.002	0.043	-0.030	-0.004

Notes: Convergent-validity check: residual CEO uncertainty regressed on firm-level political risk. Columns (1)–(2) use industry (FF12) and calendar-year fixed effects; columns (3)–(4) use firm and calendar-year fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (one-tailed for the independent variable, $\beta > 0$; two-tailed for controls).

Table 5.7: Convergent Validity: US Policy Uncertainty

	(1)	(2)	(3)	(4)
	Industry + Cal. Year FE		Firm + Cal. Year FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
US_EPU_log	0.0124** (0.0075)	0.0211*** (0.0074)	0.0123* (0.0075)	0.0239*** (0.0076)
UncQue	0.0011 (0.0035)	0.0164*** (0.0032)	0.0024 (0.0040)	0.0095*** (0.0028)
NegCall	0.0006 (0.0056)	0.1247*** (0.0089)	0.0021 (0.0073)	0.1385*** (0.0079)
lnAssets	0.0009 (0.0007)	-0.0143*** (0.0020)	0.0013 (0.0067)	0.0038 (0.0063)
TobinsQ	-0.0001 (0.0011)	0.0007 (0.0019)	-0.0000 (0.0027)	0.0007 (0.0018)
ROA	0.0281* (0.0149)	0.0367** (0.0168)	0.0677** (0.0268)	0.0467** (0.0191)
CashRatio	0.0051 (0.0087)	-0.0079 (0.0187)	0.0345 (0.0256)	0.0305 (0.0231)
DivDummy	-0.0046*** (0.0009)	-0.0074 (0.0061)	-0.0142** (0.0072)	-0.0157* (0.0083)
FirmMat	0.0023* (0.0012)	0.0009 (0.0008)	0.0005 (0.0047)	-0.0011 (0.0012)
EarnVol	0.0268* (0.0157)	-0.0074 (0.0086)	0.0353** (0.0176)	-0.0083 (0.0119)
UncPreCEO	-0.0026 (0.0032)		-0.0011 (0.0045)	
Lagged_DV	0.0202*** (0.0054)	0.5176*** (0.0131)	0.0187*** (0.0056)	0.2626*** (0.0089)
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	39,063	60,503	39,063	60,503
R^2	0.001	0.292	0.001	0.084
Adj. R^2	-0.000	0.291	-0.035	0.057

Notes: Convergent-validity check: residual CEO uncertainty regressed on United States economic policy uncertainty. Columns (1)–(2) use industry (FF12) and calendar-year fixed effects; columns (3)–(4) use firm and calendar-year fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (one-tailed for the independent variable, $\beta > 0$; two-tailed for controls).

Table 5.8: Convergent Validity: Global Policy Uncertainty

	(1)	(2)	(3)	(4)
	Industry + Cal. Year FE		Firm + Cal. Year FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
GEPU_log	0.0181** (0.0100)	0.0271*** (0.0107)	0.0187** (0.0101)	0.0309*** (0.0108)
UncQue	0.0011 (0.0035)	0.0164*** (0.0032)	0.0024 (0.0039)	0.0095*** (0.0028)
NegCall	0.0006 (0.0056)	0.1247*** (0.0089)	0.0020 (0.0072)	0.1385*** (0.0079)
lnAssets	0.0009 (0.0007)	-0.0143*** (0.0020)	0.0012 (0.0067)	0.0037 (0.0063)
TobinsQ	-0.0001 (0.0011)	0.0008 (0.0019)	0.0000 (0.0027)	0.0007 (0.0018)
ROA	0.0280* (0.0149)	0.0367** (0.0168)	0.0673** (0.0267)	0.0462** (0.0191)
CashRatio	0.0051 (0.0087)	-0.0080 (0.0188)	0.0345 (0.0256)	0.0304 (0.0231)
DivDummy	-0.0047*** (0.0009)	-0.0074 (0.0061)	-0.0142** (0.0072)	-0.0156* (0.0083)
FirmMat	0.0023* (0.0012)	0.0009 (0.0008)	0.0006 (0.0047)	-0.0011 (0.0012)
EarnVol	0.0269* (0.0157)	-0.0074 (0.0086)	0.0354** (0.0176)	-0.0082 (0.0119)
UncPreCEO	-0.0026 (0.0032)		-0.0011 (0.0045)	
Lagged_DV	0.0202*** (0.0054)	0.5176*** (0.0131)	0.0187*** (0.0056)	0.2626*** (0.0089)
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	39,063	60,503	39,063	60,503
R^2	0.001	0.292	0.001	0.084
Adj. R^2	-0.000	0.291	-0.035	0.057

Notes: Convergent-validity check: residual CEO uncertainty regressed on global economic policy uncertainty. Columns (1)–(2) use industry (FF12) and calendar-year fixed effects; columns (3)–(4) use firm and calendar-year fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (one-tailed for the independent variable, $\beta > 0$; two-tailed for controls).

Table 5.9: Analyst Cash Scrutiny: Construct Validity

	DV: CashScrutiny (% of analyst Q&A turns on cash/liquidity)			
	(1)	(2)	(3)	(4)
CashRatio	0.7530*** (0.0949)	0.8519*** (0.0967)		
HighCash			0.1754*** (0.0237)	0.1921*** (0.0236)
Leverage	—	0.2180** (0.0978)	—	0.2030** (0.0987)
lnAssets	—	-0.0562*** (0.0213)	—	-0.0744*** (0.0212)
TobinsQ	—	-0.0272*** (0.0089)	—	-0.0218** (0.0085)
ROA	—	-0.5012*** (0.1339)	—	-0.5002*** (0.1335)
Capex	—	-1.2348*** (0.2890)	—	-1.3259*** (0.2900)
DivDummy	—	-0.0674** (0.0331)	—	-0.0678** (0.0331)
sCFO	—	0.0244 (0.0274)	—	0.0291 (0.0277)
Firm FE	Yes	Yes	Yes	Yes
Cal. Year-Quarter FE	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Firms	1,868	1,853	1,868	1,853
N (calls)	75,087	72,670	75,087	72,670
R ²	0.002	0.005	0.001	0.004

Notes: Validity check that a firm's cash position predicts analyst cash scrutiny. Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (one-tailed for the cash coefficient, $\beta > 0$; two-tailed for controls).

Table 5.10: Analyst Cash Scrutiny as a Driver of CEO Uncertainty

	UncResCEO (DWZ residual)	UncAnsCEO (raw Q&A)
<i>Panel A. OLS — DV in continuous level</i>		
CashScrutiny	-0.0000 (0.0013)	0.0001 (0.0013)
<i>N</i> (calls)	41,512	41,512
Within R^2	0.000	0.000
<i>Panel B. Logit — DV = $\mathbf{1}[measure \geq median]$</i>		
CashScrutiny (log-odds)	-0.0003 (0.0074)	0.0080 (0.0085)
<i>N</i> (calls)	41,512	41,512
Pseudo R^2	0.002	0.019
Firm FE (A) / Industry FE (B)	Yes	Yes
Cal. Year-Quarter FE	Yes	Yes

Notes: Analyst cash scrutiny as a candidate driver of residual CEO uncertainty. Panel A is OLS on the continuous measure; Panel B is a logit on an above-median indicator. Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Table 5.11: Analyst Cash Scrutiny: Interaction with the Run-Up

	UncResCEO	
	(1)	(2)
CashScrutiny	0.0000 (0.0018)	0.0001 (0.0018)
PreAnnounceQtr	0.0413 *** (0.0177)	0.0439 ** (0.0189)
CashScrutiny $\times$ PreAnnounceQtr		-0.0056 (0.0111)
Controls	Yes	Yes
Firm FE	Yes	Yes
Cal. Year-Quarter FE	Yes	Yes
N (firm-quarters)	26,216	26,216
Firms	1,237	1,237
R^2	0.001	0.001

Notes: The pre-announcement run-up estimated with analyst cash scrutiny and its interaction with the pre-announcement window, on the matched universe (calls with both CashScrutiny and UncResCEO). Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (one-tailed for the directional terms, $\beta > 0$; two-tailed otherwise).

Table 5.12: CEO Uncertainty and the Bid-Ask Spread

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
			BGTLevel_Spread						BGTLevel_Spread_lead1			
ClarityCEO	-0.0477 (0.2145)	1.0617 (0.4277)	0.2122 (0.2227)	1.1056 (0.4260)	0.1644 (0.2103)	1.0547 (0.4174)	-0.1886 (0.2370)	0.5998 (0.4004)	-0.0922 (0.2399)	0.6828 (0.4055)	-0.0431 (0.2206)	0.5722 (0.3729)
UncResCEO	-0.0594 (0.1068)	-0.0687 (0.1057)	-0.0882 (0.1023)	-0.1021 (0.1007)	-0.0757 (0.0994)	-0.0893 (0.0981)	0.0825 (0.1416)	0.0696 (0.1437)	0.0742 (0.1407)	0.0545 (0.1426)	0.0984 (0.1339)	0.0751 (0.1357)
UncPreCEO	0.1664** (0.0946)	0.3496*** (0.1253)	0.0814 (0.0956)	0.2945*** (0.1229)	0.0108 (0.0919)	0.1644* (0.1184)	-0.0817 (0.0926)	-0.0288 (0.1229)	-0.1072 (0.0926)	-0.0360 (0.1233)	0.0170 (0.0872)	0.1124 (0.1186)
InAssets	-0.8039*** (0.0490)	-1.9098*** (0.2000)	-0.6339*** (0.0472)	-1.6825*** (0.2156)	-0.6352*** (0.0473)	-1.8277*** (0.2158)	-0.8218*** (0.0513)	-1.5191*** (0.1951)	-0.7857*** (0.0505)	-1.7292*** (0.2130)	-0.6568*** (0.0474)	-1.4181*** (0.2017)
TobinsQ	-0.2861*** (0.0457)	-0.4785*** (0.0900)	-0.2633*** (0.0470)	-0.5518*** (0.0963)	-0.2654*** (0.0451)	-0.5883*** (0.0956)	-0.2305*** (0.0467)	-0.3215*** (0.0850)	-0.2285*** (0.0478)	-0.4803*** (0.1002)	-0.2035*** (0.0433)	-0.4289*** (0.0935)
ROA	-9.4469*** (1.0201)	-11.6430*** (1.3867)	-5.7059*** (0.9822)	-7.6844*** (1.2654)	-5.7511*** (0.9633)	-7.8517*** (1.2490)	-9.9256*** (1.0798)	-12.8443*** (1.2992)	-8.8744*** (1.0722)	-11.9197*** (1.2732)	-7.6375*** (0.9869)	-10.5867*** (1.1921)
Leverage	0.3875* (0.2088)	1.8330*** (0.5143)	-0.1126 (0.2434)	0.6850 (0.5013)	-0.0542 (0.2264)	0.8026 (0.4954)	0.2924 (0.2224)	0.5282 (0.5516)	0.1338 (0.2248)	0.2221 (0.5596)	0.0592 (0.2111)	0.1928 (0.5258)
Capex	-0.8001 (1.1109)	0.9540 (2.9126)	-2.3622** (1.1044)	-1.5271 (2.8061)	-2.0790* (1.0845)	-1.2897 (2.7998)	-1.6463* (0.9620)	-3.2121 (2.2020)	-2.1561** (0.9823)	-4.3379* (2.2446)	-1.4770 (0.9011)	-2.4356 (2.1565)
DivDummy	0.1043 (0.0893)	0.1429 (0.2020)	0.3329*** (0.0896)	0.2549 (0.1888)	0.3156*** (0.0855)	0.2723 (0.1867)	0.0240 (0.0887)	0.1007 (0.2095)	0.1012 (0.0889)	0.1454 (0.2074)	0.1145 (0.0825)	0.1739 (0.1975)
sCFO	0.6700** (0.3405)	0.8405** (0.3472)	0.4045 (0.4604)	0.6825* (0.3789)	0.4084 (0.4126)	0.6882** (0.3375)	0.0871 (0.1759)	0.3644* (0.1877)	0.0040 (0.2174)	0.3165 (0.2055)	-0.0840 (0.1937)	0.2479 (0.1738)
Lagged_DV	0.7547*** (0.0150)	0.6569*** (0.0186)	0.7140*** (0.0167)	0.6180*** (0.0199)	0.7310*** (0.0168)	0.6318*** (0.0208)	0.7540*** (0.0156)	0.6361*** (0.0185)	0.7360*** (0.0173)	0.6135*** (0.0205)	0.7676*** (0.0162)	0.6518*** (0.0197)
StockPrice			-0.0011 (0.0014)	-0.0033 (0.0024)	0.0002 (0.0013)	0.0019 (0.0024)			-0.0001 (0.0013)	0.0101*** (0.0024)	-0.0016 (0.0012)	0.0047** (0.0022)
Turnover			-24.1012*** (1.7508)	-23.8647*** (2.2119)	-20.7840*** (1.7290)	-20.3411*** (2.1791)			-7.9396*** (1.5411)	-5.5050*** (2.0904)	-7.1303*** (1.4699)	-4.7073** (1.9633)
DailyVola			0.1229*** (0.0072)	0.1393*** (0.0081)	0.1075*** (0.0082)	0.1257*** (0.0099)			0.0404*** (0.0062)	0.0520*** (0.0068)	0.0396*** (0.0069)	0.0441*** (0.0081)
AbsSurpDec			-0.0919*** (0.0265)	-0.0241 (0.0300)	-0.0756*** (0.0253)	-0.0093 (0.0286)			-0.0050 (0.0278)	0.0119 (0.0304)	-0.0174 (0.0261)	-0.0012 (0.0285)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	42,625	42,625	42,625	42,625	42,625	42,625	43,049	43,049	43,049	43,049	43,049	43,049
R ²	0.703	0.517	0.721	0.551	0.730	0.548	0.700	0.499	0.702	0.504	0.723	0.523
Adj. R ²	0.703	0.501	0.721	0.535	0.730	0.532	0.700	0.482	0.702	0.487	0.723	0.506

Notes: Post-call bid-ask spread regressed on the three speech components across twelve specifications; the fixed-effect grid (industry or firm, by year or year-quarter) and the contemporaneous-versus-one-quarter-ahead window vary by column as indicated, and a lagged dependent variable is included. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (one-tailed for the speech components, $\beta > 0$; two-tailed for controls).

Table 5.13: Robustness: Withdrawal as a Resolution Event

	(1) UncResCEO	(2) CashRatio
PRE2 ($t-2$, pre-trend)	0.0069 (0.0179)	0.0008 (0.0024)
PRE1 ($t-1$, pre-announce)	0.0479 *** (0.0178)	0.0061 ** (0.0024)
GAP (announced, pre-close)	0.0022 (0.0187)	0.0056 (0.0034)
POST (completed)	-0.0208 (0.0127)	-0.0143 *** (0.0022)
Drop: PRE1 – GAP	0.0457 ** (0.0224)	0.0006 (0.0039)
Drop: GAP – POST	0.0230 (0.0196)	0.0199 *** (0.0041)
Drop: PRE1 – POST	0.0687 *** (0.0188)	0.0204 *** (0.0030)
<i>Controls</i>		
Leverage	-0.0430 ** (0.0212)	-0.0215 *** (0.0054)
lnAssets	-0.0063 (0.0084)	-0.0075 *** (0.0020)
TobinsQ	-0.0046 (0.0029)	0.0013 ** (0.0006)
ROA	0.0578 * (0.0325)	0.0068 (0.0061)
Capex	0.1131 (0.1051)	-0.1419 *** (0.0140)
DivDummy	-0.0110 (0.0104)	-0.0017 (0.0014)
sCFO	0.0094 ** (0.0043)	-0.0011 (0.0007)
CashRatio $_{t-1}$ (partial adj.)	—	0.7554 *** (0.0107)
Firm FE / Year-Qtr FE / Controls	Yes	Yes
N (firm-quarters)	28,191	28,191
Firms	1,321	1,321

Notes: Robustness for the matched event study with the post bin redefined to count any resolution, completion or withdrawal (quarters at and after a withdrawal enter POST). Bins, baseline, and estimator as in the matched event study. Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Table 5.14: Robustness: Static Fixed Effects without the Lagged Dependent Variable

	(1) UncResCEO	(2) CashRatio
PRE2 ($t-2$, pre-trend)	0.0068 (0.0178)	0.0052 (0.0041)
PRE1 ($t-1$, pre-announce)	0.0473 *** (0.0178)	0.0116 ** (0.0046)
GAP (announced, pre-close)	0.0018 (0.0187)	0.0128 ** (0.0057)
POST (completed)	-0.0250 * (0.0130)	-0.0189 *** (0.0053)
Drop: PRE1 – GAP	0.0455 ** (0.0225)	-0.0012 (0.0051)
Drop: GAP – POST	0.0268 (0.0199)	0.0318 *** (0.0061)
Drop: PRE1 – POST	0.0723 *** (0.0190)	0.0305 *** (0.0044)
<i>Controls</i>		
Leverage	-0.0415 * (0.0213)	-0.0529 *** (0.0189)
lnAssets	-0.0060 (0.0084)	-0.0244 *** (0.0066)
TobinsQ	-0.0045 (0.0029)	0.0101 *** (0.0019)
ROA	0.0572 * (0.0325)	0.0195 (0.0179)
Capex	0.1073 (0.1054)	-0.2534 *** (0.0396)
DivDummy	-0.0099 (0.0104)	-0.0004 (0.0045)
sCFO	0.0096 ** (0.0043)	0.0004 (0.0027)
CashRatio $_{t-1}$ (partial adj.)	—	—
Firm FE / Year-Qtr FE / Controls	Yes	Yes
N (firm-quarters)	28,102	28,102
Firms	1,320	1,320

Notes: Robustness for the cash column with static fixed effects, dropping the lagged dependent variable; the residual column is unchanged, since it carries no lag, and without the lag the coefficients describe the cash-ratio level rather than its change. Firm and calendar year-quarter fixed effects. Standard errors clustered by firm, in parentheses; significant coefficients in **bold**. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed).

Table 5.15: Construct Validity: Replication of Dzieliński et al. (2021)

	DWZ (2021)	This study	
	Table 3 (2)	Baseline	Extended
<i>Speech controls (raw frequency units; directly comparable)</i>			
CEO presentation uncertainty (<i>UncPreCEO</i>)	0.093*** (24.47)	0.089*** (14.94)	0.090*** (14.86)
Analyst-question uncertainty (<i>UncQue</i>)	0.049*** (18.18)	0.034*** (8.60)	0.034*** (8.63)
Call negativity (<i>NegCall</i>)	0.046*** (9.18)	0.068*** (8.35)	0.072*** (8.52)
<i>Firm characteristics (standardized in this study[†]; signs comparable)</i>			
Earnings-surprise decile (<i>SurpDec</i>)	0.001* (1.73)	0.003* (1.76)	0.003* (1.65)
Stock return (<i>StockRet</i>)	−0.000 (−1.59)	0.001 (0.49)	0.002 (0.76)
EPS growth (<i>EPSgrowth</i>)	0.000 (0.14)	0.001 (0.86)	0.001 (0.33)
Market return (<i>MarketRet</i>)	−0.038** (−2.52)	−0.003 (−1.24)	−0.002 (−0.89)
<i>Extended financial controls (this study only; standardized[†])</i>			
Size, B/M, leverage, ROA, current ratio, R&D, volatility	—	—	included
<i>Fixed effects and fit</i>			
CEO fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	95,296	44,900	43,638
Distinct CEOs	5,985	2,197	2,151
R^2	0.0537 [‡]	0.369	0.368

Notes: The specification is Equation 4 of Dzieliński, Wagner, and Zeckhauser (2021), reproduced variable for variable, with *ClarityCEO* the negative of the CEO fixed effect and *UncResCEO* the retained residual. *t*-statistics (in parentheses) are clustered by CEO; CEOs with fewer than five calls are excluded. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed). The samples differ: the original covers United States calls from 2003 to 2015, this study non-financial, non-utility United States calls from 2002 to 2018. [†]Firm-characteristic controls are standardized here, so their magnitudes are not comparable to the original's raw-unit coefficients, though signs and significance are. [‡]The original reports R^2 as the incremental power of the controls over a CEO-fixed-effects benchmark of 0.31 (about 0.364 in total); this study reports total R^2 inclusive of the absorbed fixed effects.

Appendix I Cash-Scrutiny Measure: Definition and Word Lists

Variable definitions.

- **CashScrutiny_{*i,t*}** — the share of call *i*'s analyst question-and-answer turns whose text contains at least one *cash-level* or *liquidity* term (listed below); a turn-level indicator averaged over the call's analyst Q&A turns.
- **HighCashScrutiny_{*i,t*}** — an indicator equal to one when CashScrutiny_{*i,t*} exceeds its sample median; because the median call draws no cash scrutiny, it flags calls with any cash scrutiny at all.

Search terms. An analyst Q&A turn is flagged if, after the *excluded* phrases are blanked out, its lower-cased text contains any phrase below, matched on a word boundary (case-insensitive). Every literal spelling actually searched is listed: where a phrase has a singular/plural or hyphenated/closed variant (e.g. 'noncash' and 'non-cash'), both forms appear. Disposition/payout terms (dividends, buybacks) are deliberately *not* used here: they load on firm size and maturity and would correlate with the cash position spuriously rather than through cash attention.

- **Cash level** (20 forms) — *why*: they name the firm's cash stockpile or balance itself, the direct object of attention to how much cash the firm holds.
'cash holding', 'cash holdings', 'cash balance', 'cash balances', 'cash position', 'cash on hand', 'cash on the balance sheet', 'cash reserve', 'cash reserves', 'cash and cash equivalents', 'cash and equivalents', 'cash and shortterm investments', 'cash and short-term investments', 'net cash', 'cash pile', 'cash hoard', 'cash stockpile', 'excess cash', 'idle cash', 'surplus cash'
- **Liquidity** (5 forms) — *why*: standard analyst synonyms for the same liquid-asset buffer.
'liquidity', 'liquid assets', 'shortterm investments', 'short-term investments', 'marketable securities'
- **Excluded** (17 forms) — *why*: they contain "cash" but denote cash *flow*, earnings, an accounting basis, or an idiom rather than the cash holdings, so they are blanked out before matching to prevent false positives.
'free cash flow', 'operating cash flow', 'cash flow statement', 'cash

flow', 'cash flows', 'cash conversion cycle', 'cash conversion', 'cash basis', 'cash cow', 'cash compensation', 'noncash', 'non-cash', 'cash register', 'cash crop', 'cash taxes', 'cash earnings', 'cash in on'

Appendix II Variable Definitions

The cash ratio and the firm-financial controls $X_{i,t}$ are constructed from quarterly Compustat North America fundamentals as follows.

Variable	Definition	Interpretation
CashRatio	cheq/atq	cash and equivalents over total assets
<i>Firm-financial controls $X_{i,t}$</i>		
Leverage	$(\text{dlcq} + \text{dlttq})/\text{atq}$	interest-bearing debt over assets
lnAssets	$\ln(\text{atq})$	firm size
Tobin's Q	$(\text{atq} + \text{cshoq} \cdot \text{prccq} - \text{ceqq})/\text{atq}$	market-to-book
ROA	iby/assets	profitability
Capex	capxy/atq	investment intensity
DivDummy	$\mathbf{1}[\text{dvy} > 0]$	dividend payer
sCFO	five-year rolling standard deviation of $\text{oancfy}/\text{atq}_{t-1}$	cash-flow volatility